FMI	ADMINISTRATIVE INFORMATION	FMI No: 25400

FMI Title: VCT Global Table Belt Tension Adjustment

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☑ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☑ **Staged FMI** — **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.

☑ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.

☐ **Parts Ordering**: GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator

☐ **No Parts** are required to complete this FMI.

☐ **Batch with** ______________________________

To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).

☐ **Non-Staged** — **No Customer List is provided**. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center 800-321-7937 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:

Unit Tracked: All VCT Systems

FMI Instructions Part No.: 5220940-100 Rev 3

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS - Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25400 FMI Title: VCT Global Table Belt Tension Adjustment

Customer Full Name:

Number and Street Address:

City and Country:

Country Code Site Number System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer Comments

Revision History

Revision	Date	Reason for Change
1	4/25/2007	Initial Release
2	7/06/07	Incorporated changes based on pilot feedback
3	10/04/07	Add DVCT to Effectivity

Effectivity

The FMI effectivity includes CT VCT64, VCT32, VCTSTD, and PET DVCT systems. The following are tracked gantry model numbers

Gantry Model Numbers:

- 5124069
- 5124069-2
- 5124069-5
- 5124069-6
- 5126093
- 5158924
- 5178168

Purpose

The purpose of this FMI is:

- To check interference of Detector DHC and CFC cables to the stationary part.
- Check detector plenum eyebolts for proper torque.
- To tension the following Global Table belts to the required frequency specification: IMS drive belt, cradle motor belt, and cradle drive belt. A laptop belt tension software tool will be used to assist in setting belt frequency. ***The Volume Shuttle feature (a purchased option) requires belts to be tensioned within the required frequency specification in order to function properly.***

All systems require the Detector Cables Check and Detector Plenum Eyebolt Torque Check.

Very recently installed Tables may not require the Belt Tensioning Adjustment. Check below to see if your table is affected. Note, tracking for the FMI is based on gantry model number to ensure all systems are covered. The cut-in date for the belt tensioning is based on the table serial number.

If your table serial number ends with .YCx - your table requires Belt Tensioning Adjustment.

If your table serial number ends with ...YMx - your table requires Belt Tensioning Adjustment.

If your table serial number ends with ... HMx - check the table below:

Belt Tension Adjustment not needed if you have the following:			
If you have Table Model	**With Table Model #**	**And Serial # is greater than this:**	**And Mfg Date is later than this:**
VT2000	5121647	161010HM4	05/11/2007
VT1700	5122080	161874HM3	05/14/2007

Prerequisite FMIs

None

Related FMIs

FMI 25394: 07MW18.4 VCT Quality Release

FMI 25400 is a prerequisite to FMI 25394. Table Belt Tension Adjustment is needed for the Volume Shuttle option introduced with 07MW18.4 VCT Release.

FMI 26809: IMS Bearing Block Replacement

If FMI 26809 has been completed prior to FMI 25400, section 2, ‘IMS Belt Tension Procedure’ can be omitted.

Furnished Materials

FMI 25400, Kit Part Number 5220940		
Qty	**Description**	**Part #**
1	FMI 25400 Document	5220940-100
1	Microphone	5224640
1	Belt Frequency CD	5220799

Tool Required

- Standard FE tool kit. Specific tools needed for this FMI are:
  - Metric Allen Wrench Set
  - 4mm Allen Socket
  - Phillips Screwdriver
  - Long Flat Head Screwdriver
- A functioning Laptop computer with working sound card is also needed to complete this FMI.

Personnel Requirements

1 FE

Labor: 2 hours,

Travel: 1 hour

Procedure

Detector Cables – Check for Interference to Stationary Part

It was found on a few systems that the detector DHC and CFC cables contacted the stationary part of the gantry during rotation, making some tapping noise. If Service Note SN06120807a has not been completed, please reference the Service Note in Appendix B.

Detector Plenum Eyebolt Torque Check

VCT detector plenum Eyebolts may not have had proper torque applied during manufacturing. There are two Eyebolts, one on each side of the air plenum.

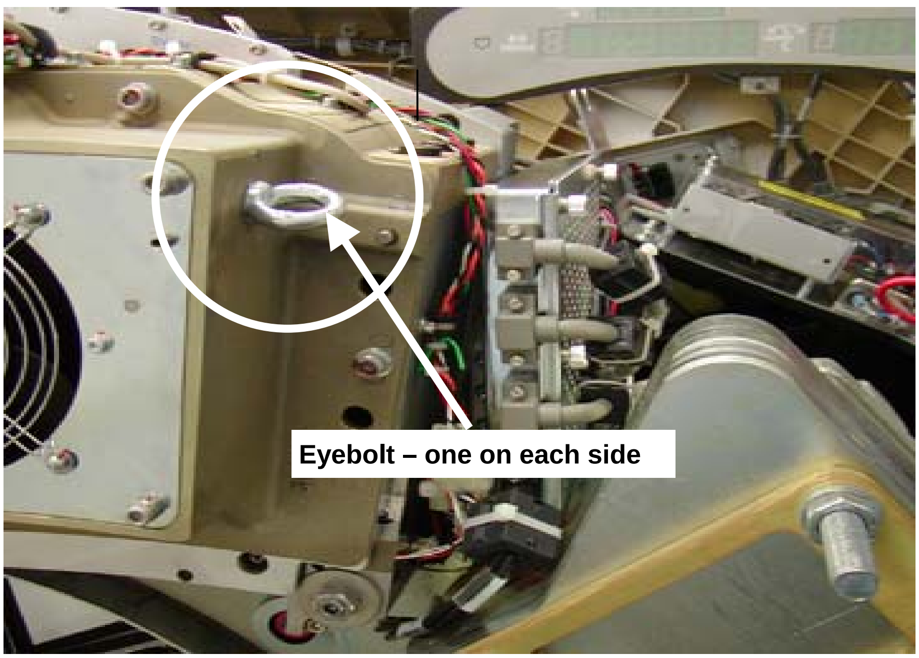


Figure 1 - Eyebolt Example

Engineering analysis determined very low risk of any Eyebolt coming out of the plenum. To ensure that Eyebolts do not come loose over time (over the life of the product), check the torque and tighten, if necessary, by following the following steps:

1. Ensure the gantry right side cover removed and Axial Drive & HVDC are disabled.
2. Position the Gantry for access to the detector plenum for one of the Eyebolts.
3. Lock gantry rotation.
4. Using just your hand, check the Eyebolt to see if you can loosen the Eyebolt. Do not use excessive force, if the Eyebolt was not properly installed, it will be obviously loose by use of minimal hand force. **DO NOT remove the Eyebolt if loose.**

5. If Eyebolt is loose perform this step.

   a. Turn the Eyebolt by hand clockwise until the shoulder of the eyebolt is firmly seated against the plenum.

   b. With a screw driver, insert the shaft through the Eyebolt. Turn the eyebolt clockwise more than 1/8 turn and no more than 1/4 turn. This will ensure proper torque is applied to the eyebolt. See Figures 2 and 3.

   c. With a marker, draw a line from the plenum onto the shoulder of the Eyebolt. This can be used to make sure the eyebolt was tightened and to see if anything changes in the future.

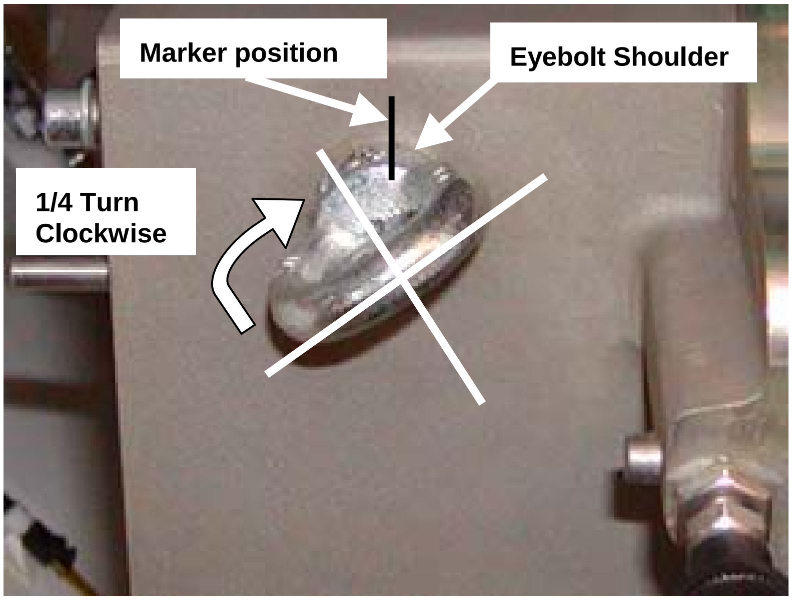


Figure 2 - Torque rotation (no more than ¼ turn)

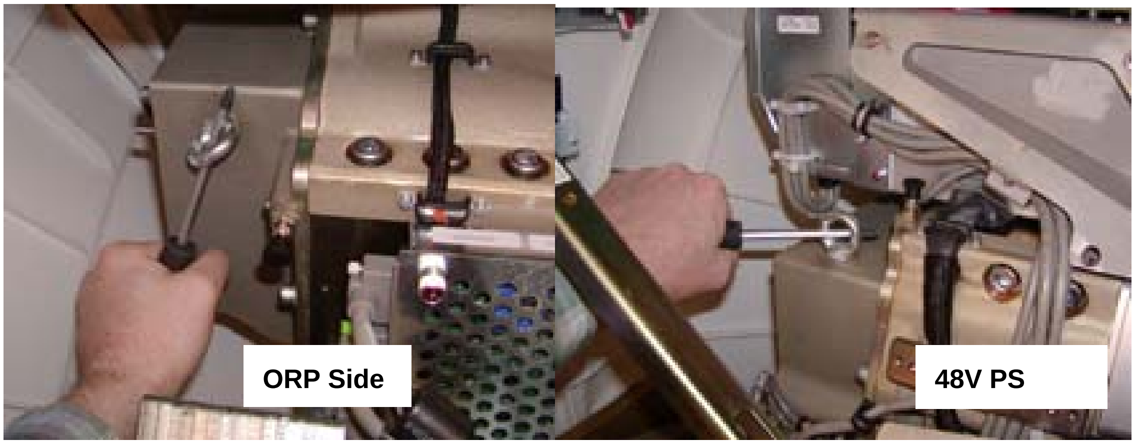


Figure 3 - Screwdriver applied Torque

6. Repeat steps 2 through 5 for the Eyebolt on the other end of the detector plenum.

Loosen gantry rotation lock and turn on the 120VAC service switch to aid in moving gantry to access the other Eyebolt as needed.

7. When done with both eyebolts perform the following steps:
    a. Loosen gantry rotation lock
    b. Turn on the 120VAC service switch
    c. Turn on HVDC and Axial drive service switches
    d. Press the table drive enable push button
    e. Install gantry side cover

Belt Frequency Adjustment

1- Laptop Setup for Belt Frequency Tool

1. Plug the microphone provided with this FMI (part # 5224640) into your laptop microphone port. The microphone port location will vary depending on the type of laptop. **Make sure to plug into the microphone port and not the headphones port (see picture below).**

Figure 4 – Microphone Port

2. For Windows XP:

On your laptop select:

Start -> **Control Panel** -> open > ***<u>Sounds and Audio Devices</u>*** >

(If your laptop has Windows 2000 see Appendix A for set up)

3. Select the "**volume"** tab on the **Sounds and Audio Devices Properties** window. Then select the "**Advanced**" button shown in the red box below.

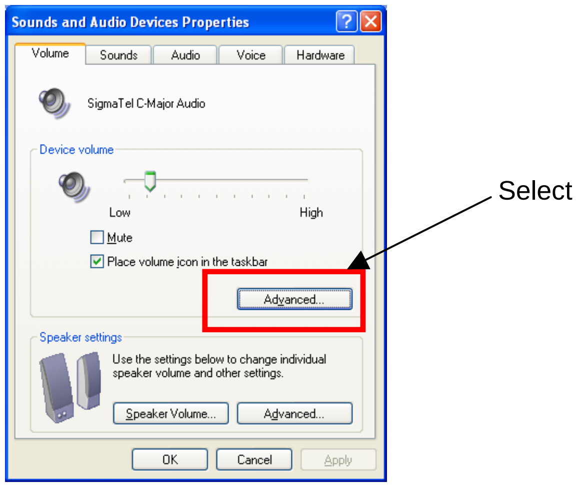


Figure 5 – Volume Tab

4. Select **Options**, then **Properties**.

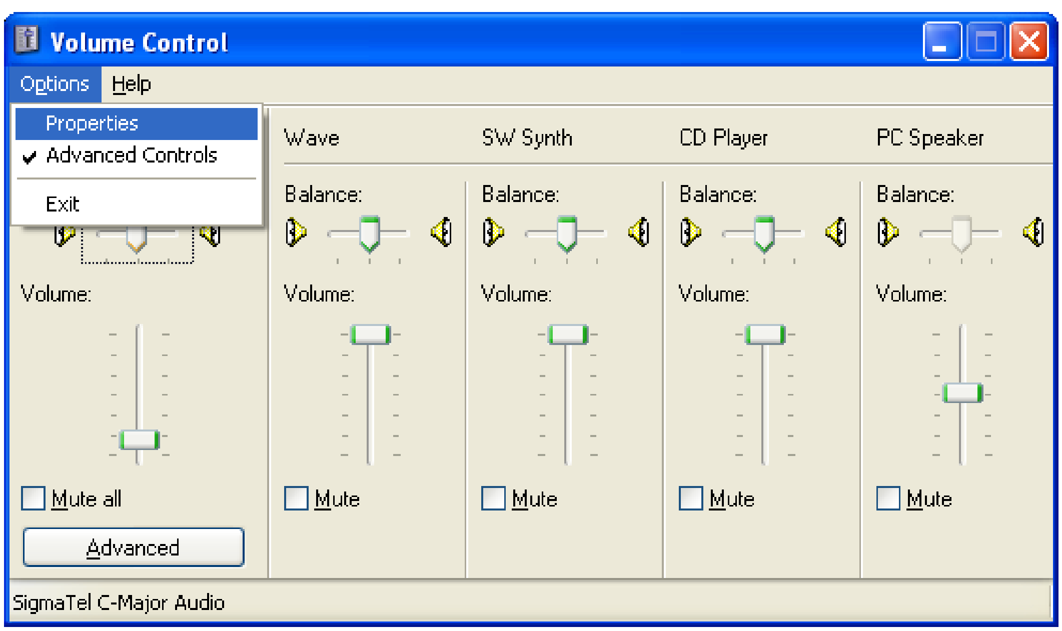


Figure 6 – Volume Control

5. Select **Recording** and **Microphone**. Make sure Stereo Mix is not selected. Then select **OK**.

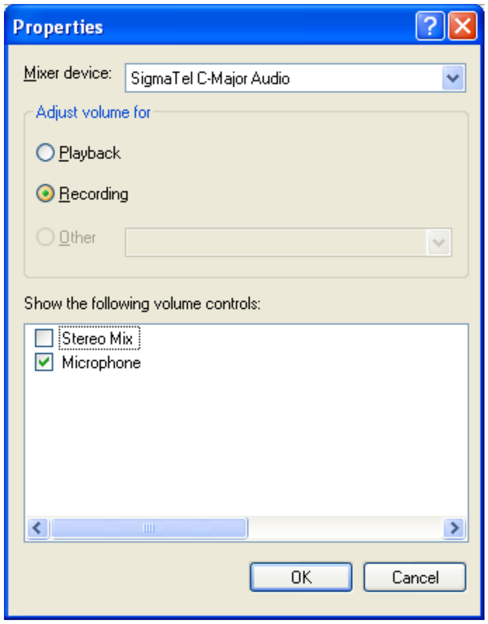


Figure 7 – Selecting Microphone

6. Set the microphone Volume to approximately 25% (It is very important to match the picture below as close as possible in setting the volume level) and be sure that the Select box has a checkmark. You can leave the microphone window open on your desktop.

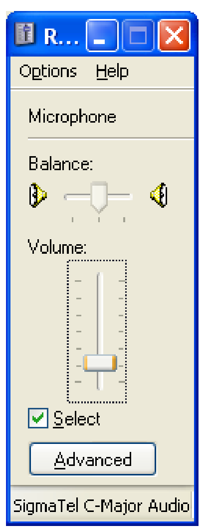


Figure 8 – Volume Adjustment

7. Load the Belt Frequency CD part #5220799 into your laptop CD drive. This will automatically load the program for measuring Belt Frequency and run it in your PC (see picture below). The name of the program file is **Bfreq1_4.exe (Version 1.4)**. The program also will be automatically stored on your laptop desktop. You will not need the CD to run the program in the future since you can just double click the file on your desktop.

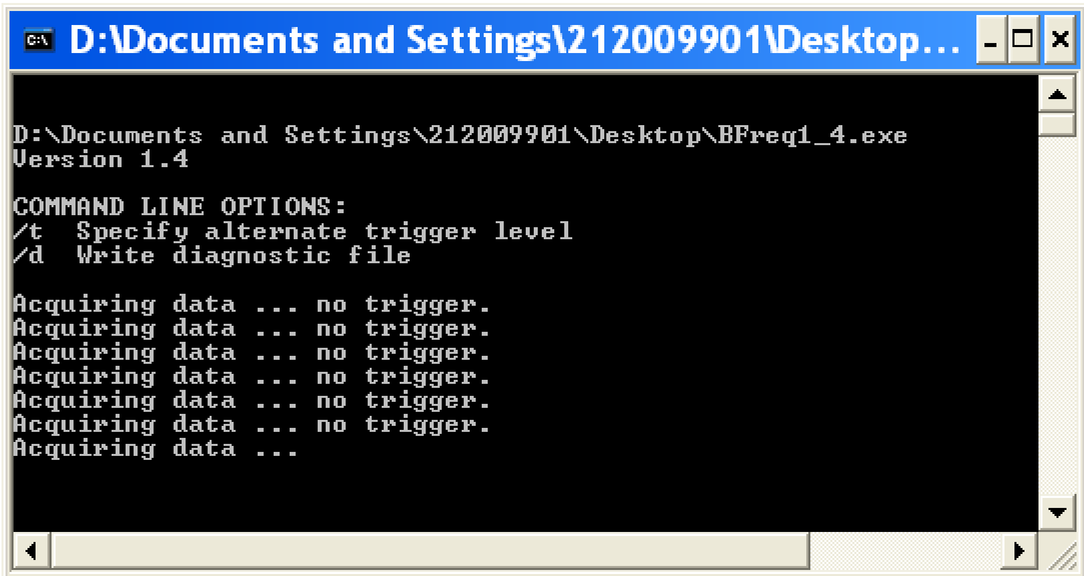


Figure 9 – Bfreq1_4.exe Program

8. The program records data from the sound card in three-second blocks. A new Acquiring data line will appear every three seconds.

   Checking the operation of the microphone:

   Talk into the microphone. The program will show 'OVERLOAD' or a frequency value if the microphone is set up correctly and your sound card is working properly.

9. This step will describe how to use the laptop tool and microphone to measure belt frequency. Read completely and refer to this step as needed when adjusting belt tensions.

   By holding the microphone close to the belt and strumming the belt one time per time capture block, the program will record the first vibration mode frequency of the belt. The frequency that is recorded can be compared to the frequency specification for the belt.

   a. It is important to use this tool in a quiet location. **Minimize external noise as much as possible.**

   b. It is important to hold the microphone by the handle and not touch the actual microphone during measurements. Holding the actual microphone will cause finger-generated noise that can fool the program.

   c. Hold the microphone 2 – 3 cm (**about an inch**) from the center of the belt span being measured. Do not hit the microphone with your fingers or bump the microphone into anything when strumming the belt.

   d. Strum the belt with your fingertip or fingernail. The belt should be released sharply like when strumming a guitar string. Generally the belt needs to be displaced about 1mm (**less than 1/16th of an inch**) or less from rest position when strumming it.

e. Strum the belt one time immediately after the program writes a new “Acquiring data” line. It is important to take the measurement immediately so that the program can record the entire belt vibration within the three second block.

f. If after strumming a belt “no trigger” is displayed after block capture, either move the microphone closer or increase the amount that the belt is displaced from the rest position (i.e. increase the intensity). And similarly if after plucking a belt “OVERLOAD” is displayed after block capture, either move the microphone further from the belt or decrease the amount that the belt is displaced from the rest position when plucked.

g. Here and there you will get some false/odd readings. Ignore them. It is probably due to background noises. Continue to take measurements until consistent results are achieved. In a quiet area with careful control, results will be very repeatable.

Note:

For additional information on the usage of the belt frequency tool, please see Appendix C “Operation of Belt Frequency Tool”

2- IMS Belt Tension Procedure

1. Raise the table to maximum height.
2. Move the cradle and IMS to out limit position.
3. Remove power from the table by turning off 120VAC, Axial Drive and HVDC switches on Service Switch Panel.
4. **If IMS Block Replacement FMI 26809 has been recently completed on this system, you can skip the following IMS belt tension procedure and go to section 3, Cradle Motor Drive and Cradle Drive Tension Procedure. If you are not sure, then continue with step 5.**
5. Remove Top Covers (Right/Left) and the IMS Rear Cover and IMS Side Covers (Figure 17).

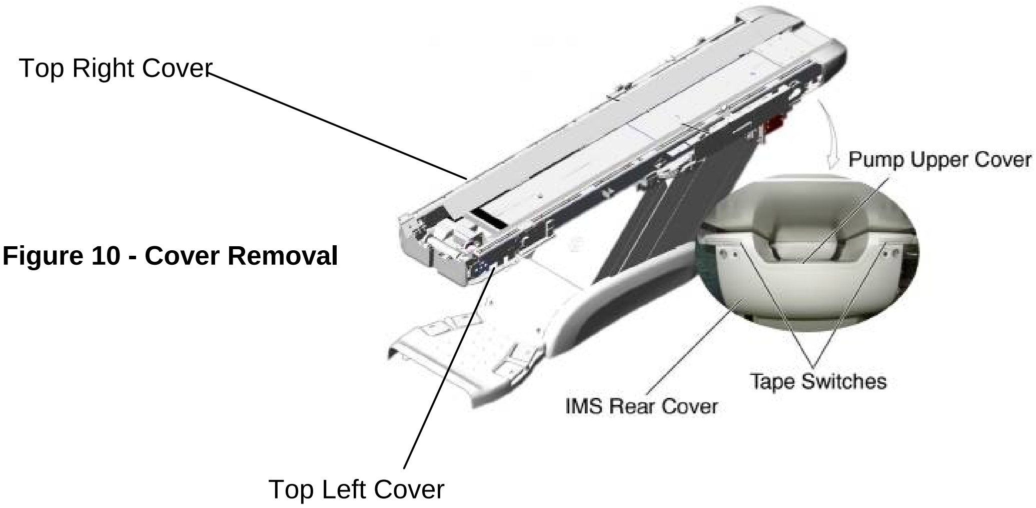


Figure 10 - Cover Removal

6. Disconnect the cable connector of the tape switch. Remove the pump upper cover as follows:
   a) To make re-installation easier, mark the position of pump upper cover on both sides by Outlining the edge with a marker or pencil.
   b) Disconnect the cable connectors of the tape switches and touch sensors.
   c) Remove six (6) screws holding the both sides of the pump upper cover, and remove it.

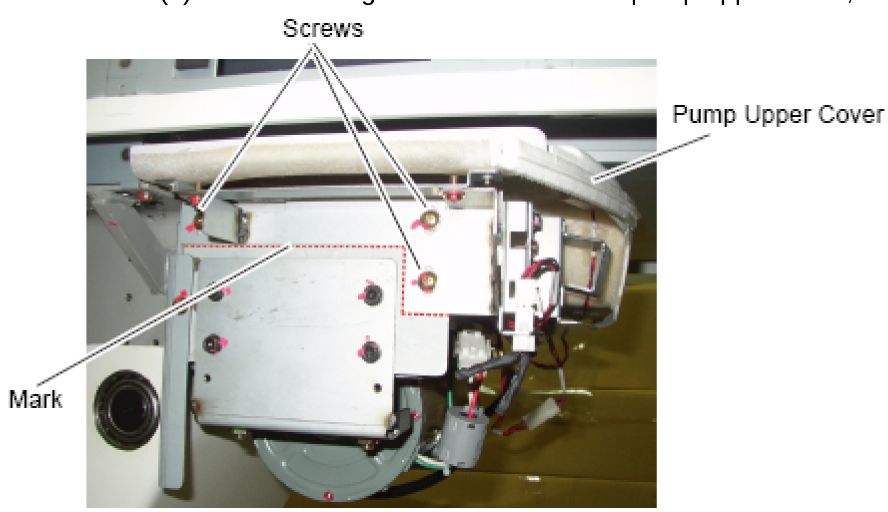


Figure 11 - Pump Cover Removal

7. Remove IMS motor cover exposing IMS Drive Belt.

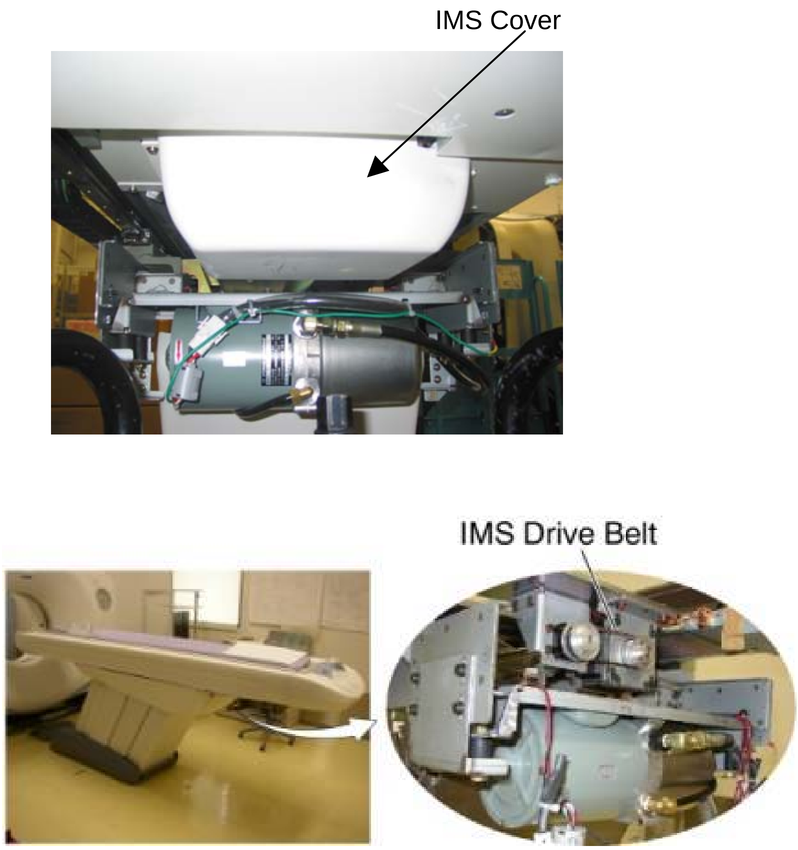


Figure 12 - IMS Motor Cover, Drive Belt

8. Set up laptop such that the monitor is visible while adjusting the belt tension.

9. If the clean damper is mounted on the IMS Motor Shaft, it will need to be removed to access the IMS Drive Belt. Remove the clean damper by loosening the two setscrews on the backside of the clean damper. Note the orientation of the setscrews on the shaft for replacement.

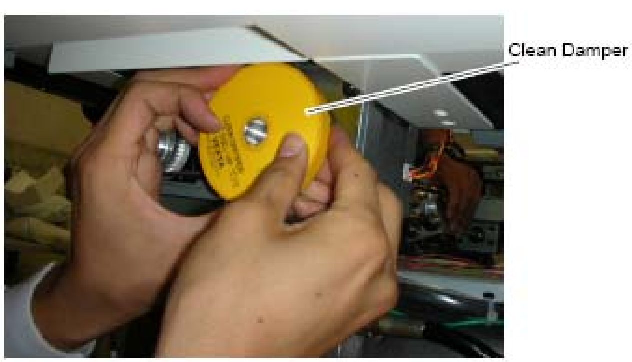


Figure 13 - Clean Damper

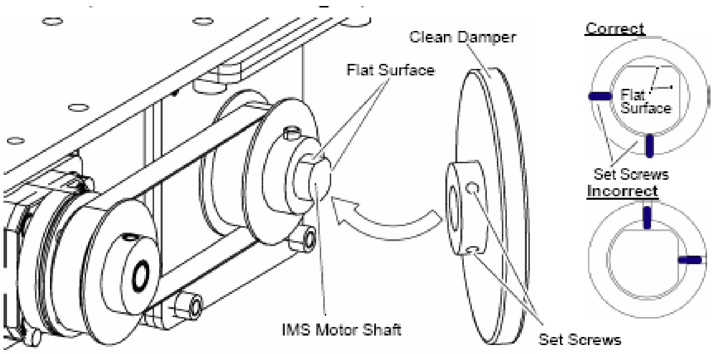


Figure 14 - Clean Damper Installation

10. Position microphone and lightly strum the top of the IMS belt in the middle of the belt span to make it vibrate. Measure the frequency using the laptop belt frequency tool and procedure.

If reading is 'overload', do not strum so hard. If reading is 'no trigger', strum a little harder. Moving the microphone a bit closer or a bit farther can also help in these situations. Continue until you get at least 3 consecutive measurements that are within 5 HZ of each other.

Adjust Belt Tension in small increments. If the frequency is high, loosen the belt. If the frequency is low, tighten the belt.

Figure 15 – IMS Belt Tension Frequency Measurement

11. Compare measurement value to the IMS belt frequency spec of **(Between 100 and 140 HZ).**

12. If frequency is out of spec, proceed to step 13. If within spec, proceed to Section 3 'Cradle Motor and Cradle Drive Belt Tension Procedure'.

13. Loosen 2 screws on IMS motor bracket. Tighten belt by using flat head screwdriver to push motor bracket assembly. Tighten 2 screws securely while keeping tension on the belt. (For torque reference, see Appendix D)

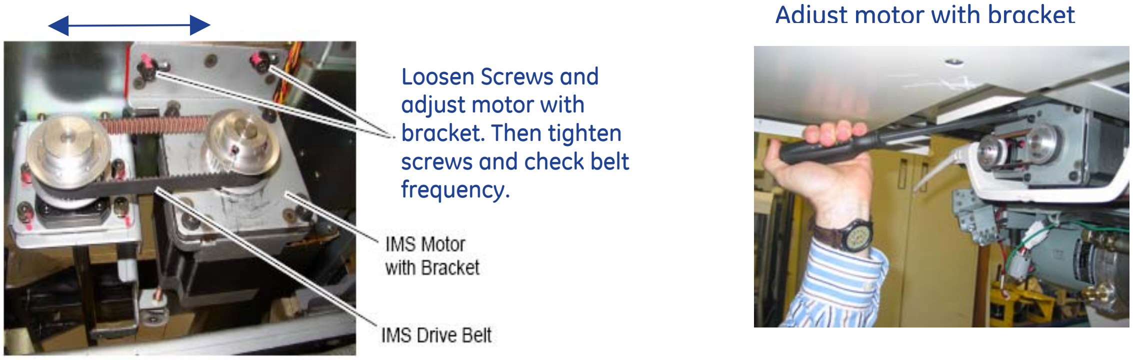


Figure 16 – IMS Belt Tension Adjustment

14. Continue to adjust belt and measure frequency until it is within spec.
15. If you removed the clean damper earlier in the procedure, please re-install it on the IMS Motor Shaft. Please refer back to step 9 for proper installation.
16. IMS belt adjustments are complete at this time. IMS covers can be re-installed at this point or can be installed at the completion of the FMI.

3- Cradle Motor and Cradle Drive Belt Tension Procedure

1. Ensure the table is at maximum height.
2. Ensure the cradle and IMS are at out limit position.
3. Ensure 120VAC, Axial Drive and HVDC switches on Service Switch Panel are off.
4. Ensure Top Covers (Right/Left) are removed. Remove Cradle, Tray-F, Front Top Cover, and Front Left Rail Cover.

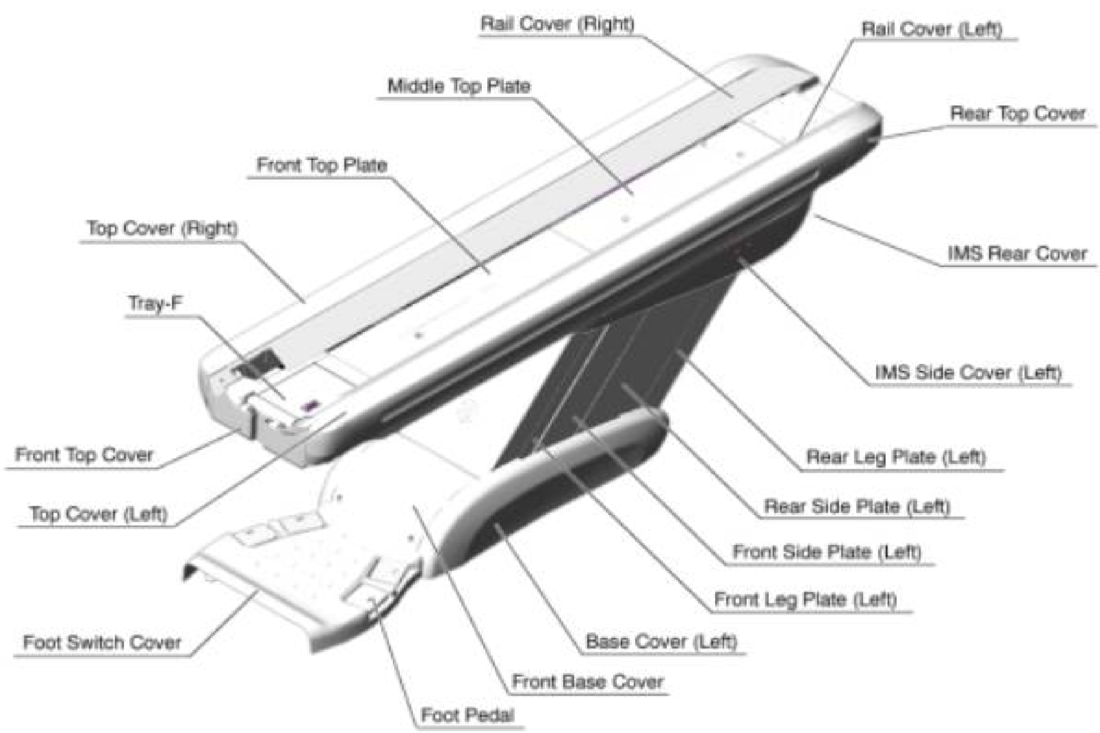


Figure 17 – Table Parts

Note: Cradle Drive Belts will be exposed. See picture below.

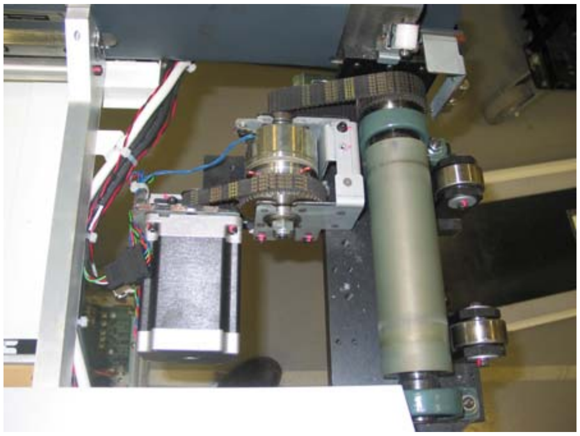


Figure 18 – Cradle Drive Belts

Cradle Motor Belt

5. Inspect verify that the pulley (or Lock Pin) on the Cradle Motor Clutch Assembly is not coming off the clutch shaft (previously inspected in FMI 26808).

If the pulley is not secure, order 5127493 (Cradle Motor Clutch Assembly) and replace the defective assembly. Complete a PQR for this problem (to help determine the number of problem sites).

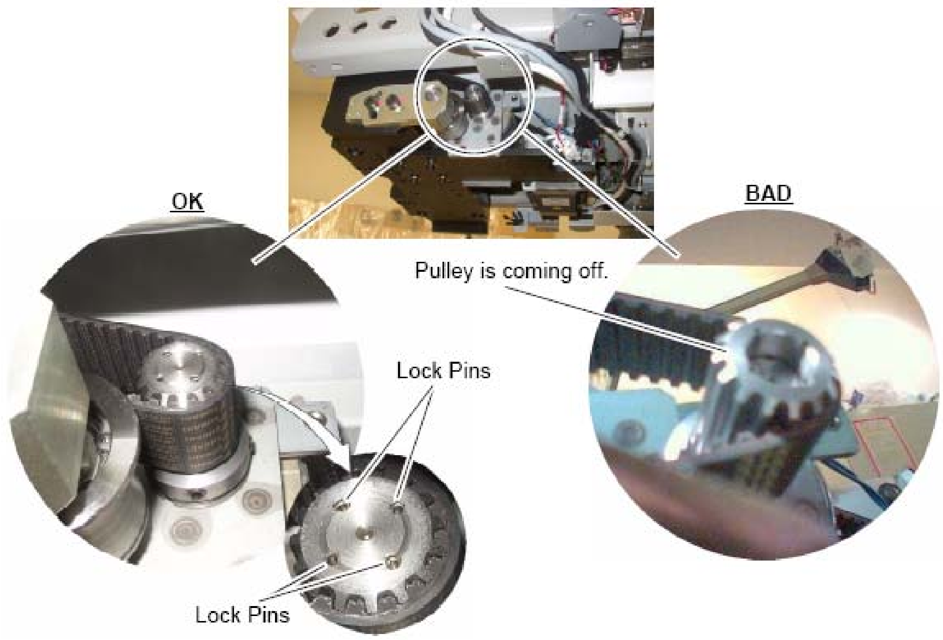
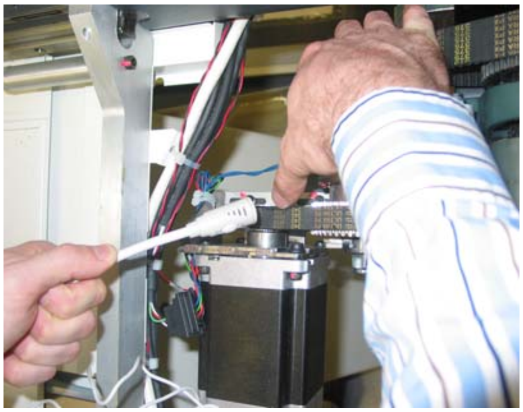


Figure 19 Pulley of the Cradle Motor Clutch Assembly

6. Set up laptop such that the monitor is visible while adjusting the belt tension.

7. Position microphone and lightly strum the top of the cradle motor belt in the middle of the belt span to make it vibrate. This is the narrower of the two cradle belts. Measure the frequency using the laptop belt frequency tool and procedure.



If reading is 'overload', do not strum so hard. If reading is 'no trigger', strum a little harder. Moving the microphone a bit closer or a bit farther can also help in these situations. Continue until you get at least 3 consecutive measurements that are within 5 HZ of each other.

Adjust Belt Tension in small increments. If the frequency is high, loosen the belt. If the frequency is low, tighten the belt.

Figure 20 – Cradle Motor Belt Frequency Measurement

8. Compare measurement value to the cradle motor belt frequency spec of **(Between 311 and 355 HZ).**
9. If frequency is out of spec, proceed to step 10. If it is within spec, go to Cradle Drive Belt Tension procedure step 13.
10. For cradle motor belt adjustment, loosen 3 bolts shown below.

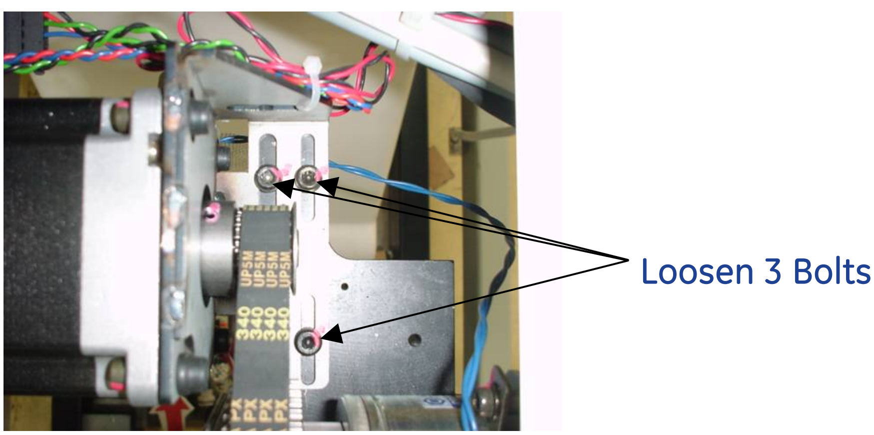


FIGURE 21 – CRADLE MOTOR BELT ADJUSTMENT BOLTS

11. Adjust belt using flat head screwdriver to slide the motor bracket and then tighten the three bolts securely while keeping tension on the belt. (For torque reference, see Appendix D)

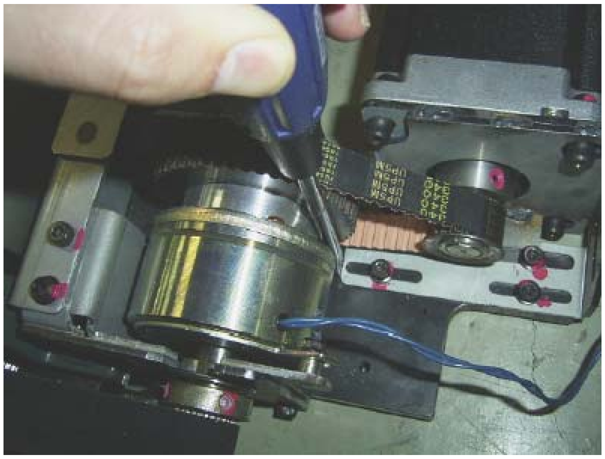
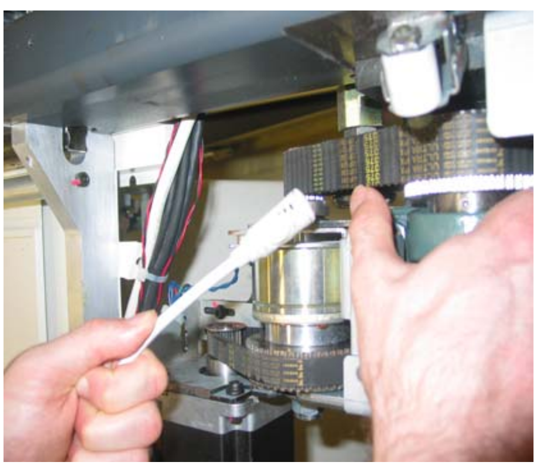


Figure 22 – Cradle Motor Belt Adjustment

12. Continue to adjust belt and measure frequency until it is within spec. Then go to step 13 for cradle drive belt.

Cradle Drive Belt

13. Set up laptop such that the monitor is visible while adjusting the belt tension.
14. Position microphone and lightly strum the top of the cradle drive belt in the middle of the belt span to make it vibrate. This is the wider of the two cradle belts. Measure the frequency using the belt frequency tool and procedure.



If reading is ‘overload’, do not strum so hard. If reading is ‘no trigger’, strum a little harder. Moving the microphone a bit closer or a bit farther can also help in these situations. Continue until you get at least 3 consecutive measurements that are within 5 HZ of each other.

Adjust Belt Tension in small increments. If the frequency is high, loosen the belt. If the frequency is low, tighten the belt.

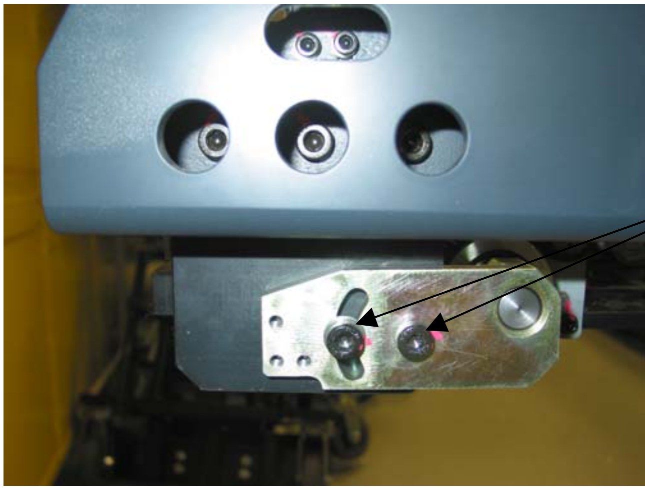
Figure 23 – Cradle Drive Belt Frequency Measurement

15. Compare measurement value to the cradle drive belt frequency spec of **(Between 222 and 246 HZ).**

16. If frequency is within spec, reassemble all the covers and verify that all table functions are operating normally. If frequency is out of spec proceed to step 17.

17. For the cradle drive belt adjustment, loosen 2 bolts on the cradle drive belt tensioner to remove tension on the belt. The bolts are located on the left side of the IMS.

SIDE VIEW



Loosen 2 Bolts

Figure 24 – Cradle Drive Belt Adjustment Bolts

18. ***Adjusting Cradle Drive Belt Tension:***

Adjust belt using flat head screwdriver to set the tension on the cradle drive belt. Tighten the two bolts securely while keeping force applied to the screwdriver. (For torque reference, see Appendix D)

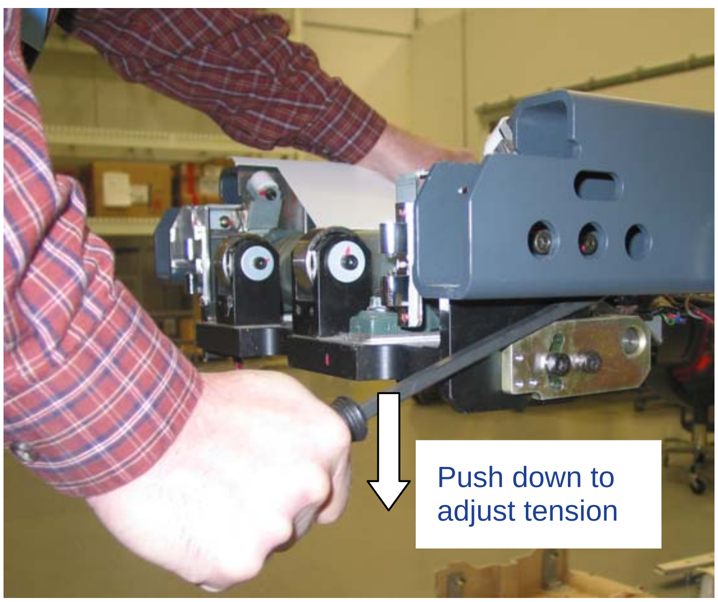


Figure 25 – Cradle Drive Belt Adjustment

19. Continue to adjust belt and measure frequency until it is within spec.

20. Reassemble all covers except the Front Top Cover" (The Front Top Cover will need to stay off so that Cradle Motor Clutch Assembly can be inspected after the belt is tightened and the IMS / Cradle has been exercised)

Finalization

1. Power up the table from the service Switch Panel in gantry.
2. Move the cradle and IMS in and out completely 10 cycles. Raise and lower the table a couple times and verify that table movement is operating normally.
3. From the bottom side of table (where the Front Top Cover would be installed) look at the cradle motor belt pulley. Go to step 3.5 and verify that the pulley is in the correct position. If the pulley is moving off the shaft, follow the instructions in step 5 to replace. If any parts are replaced, the belt frequency adjustment will need to be re-done.
4. Re-install Front Top Cover and ensure that all other covers are in place.

Finishing Up

1. Perform an application scan to ensure proper system operation before turning the system over to the customer.
2. Close FMI per regional procedure. Remove FMI document from customer site.

Appendix A

Laptop Setup for Belt Frequency Tool

1. On you laptop select: **Start** -> **Control Pane**l **->** open -> ***Sounds and Multimedia Devices*** >
2. Select the "***Audio***" tab on the top of the form
3. Select < ***sound recording*** >, select "***Volume***" tab
4. On the *Volume* Control select < *Options* > then < *Properties* >
5. In the Adjust *Volume* for, select the "***Recording***" button and make sure the box next to Microphone is "***checked***"
6. Select **OK** button
7. Be sure the Volume is turned to 25% and the Select box is "***checked***"
8. You can keep this window open
9. Return to Section 1, Step 7

Appendix B

VCT Detector Cable Interference to Stationary Part

Service Note 06120807a

Class C information is Property of GE.

Date: December 8, 2006

Subject: VCT Detector Cable Interference to Stationary Part

Effectivity: VCT (7.x systems) with 2006 Style Plenum
(This includes 64-slice Sherlock detector, 64-slice HALO detector, and 32-slice detector.)

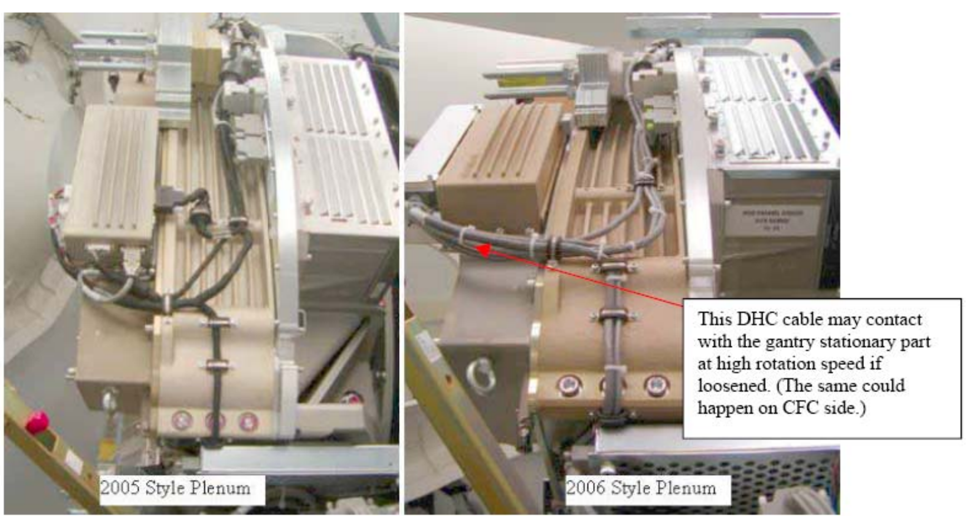
Issue:
It was found at a few systems in manufacturing and field that the detector DHC and CFC cables contacted the stationary part of the gantry during the rotation, making some tapping noise. The cause of the problem is inappropriate clamping of the cables on the 2006 style plenum. This problem does not exist on the 2005 style plenum.

Resolution:
If unusual noise is heard from the gantry during the rotation, check the DHC/CFC cables according to the procedure below. It is also recommended to perform this check during the PM as a preventive action. (The same check will be included in a future FMI.)

1) Remove gantry right side cover.

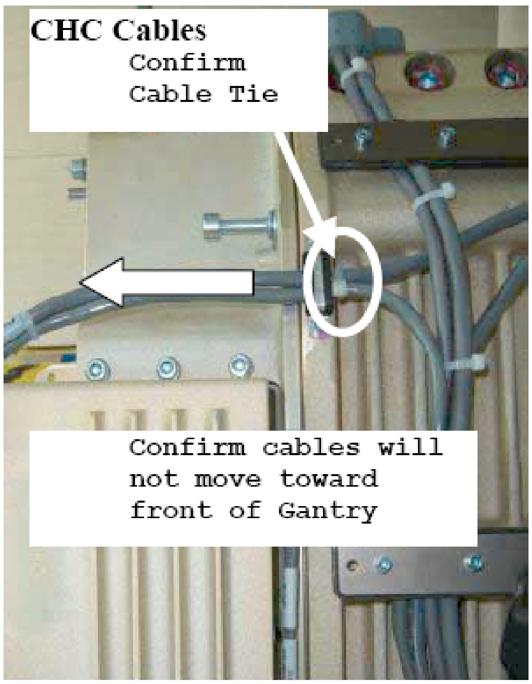
2) Disable Axial Drive and HVDC from the service switch panel.

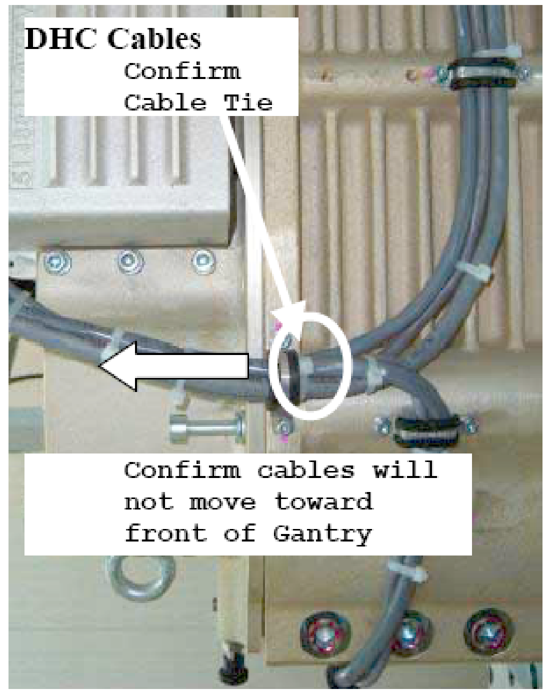
3) Rotate the gantry to see the detector plenum.
   - If plenum is old 2005 style (5117776) Shown below, go to step 6, no change required.
   - If plenum is 2006 style 5158921or 5169298 shown below, got to step 4.



4) If there are zip ties (or ty-raps) just behind the cable clamps at both DHC side and CFC side as shown below, go to step 5. Install zip ties if they are not there.

NOTE: Use a zip tie gun if available to prevent over tightening of zip tie. Do NOT over tighten zip ties as cable damage will occur.





5) Confirm the cables cannot **be moved toward the front of the gantry** as shown by the arrow in the figures above. If cables can move, tighten the zip tie or add another as needed.

6) Enable HVDC and Axial Drive.

7) Reinstall the gantry side cover.

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Appendix C
Operation of Belt Frequency Tool

A sound card and compatible microphone are used to sense the oscillating pressure around a freely vibrating belt. Plucking the belt with a finger initiates the free vibration. The frequency of free oscillation is related to the tension in the belt, the belt span, and mass density of the belt material. This is the same relationship governing the vibration frequency (pitch) of piano and guitar strings as well.

The belt frequency tool captures 3 seconds of recorded data surrounding a belt pluck. The tool uses an algorithm to find the max amplitude signal within the 3 seconds of data for the filtered hi/low amplitude range. The program reports the frequency of that signal. The max amplitude signal should be from the first vibration mode of the belt (where belt span equals 1/2 wavelength). By finding the max amplitude signal within the three seconds of data, this helps to eliminate background noise fooling the program (this is better than reporting the first signal recorded which may be background noise if you do not pluck at the very beginning of the 3 seconds). Plucking the belt harder or softer does not change the first vibration mode frequency. The program is also set up to filter very high or very low amplitude signals. If you pluck too soft, no trigger will be reported. If you pluck too hard, overload will be reported. This amplitude filtering also helps to eliminate background noise. You must calibrate pluck to be within the correct amplitude range. Once you pluck within the correct amplitude range consistently, you will report consistent first mode frequency for the belt span which can be compared against the frequency specification.

This tool is very similar to other commercially available belt tension measurement devices. It uses similar operating principles of recording sound and reporting back the first vibration mode frequency of the belt. The tool was verified by setting the following belts to their frequency spec: cradle drive 15mm, cradle drive 25mm, IMS drive, elevation POT, gantry axial drive. After the belt was set to spec using the tool, it was measured 10 times using a Gates sonic meter (commercially available tool to measure belt frequency). The belt was then measured 10 times using the software tool and this was repeated for a total of three different operators. Each operator used a different laptop. The frequency results for the tool in all cases was within a couple hz of the results for the Gates sonic meter.

Appendix D
Torque Reference

Belt	Torque Newton Meters	Torque Foot Pounds
IMS Belt	11.5	8.5
Cradle Motor and Clutch Belt	6.9	5.1
Cradle Drive Belt	17.2	12.7